FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – I
PAPER –2
TEST DATE: 28-12-2022

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B**.

Section – A *(01 – 04, 19 – 22, 37 – 40)*: This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A *(05 –10, 23 – 28, 41 – 46)*: This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – B *(11 – 18, 29 – 36, 47 – 54)*: This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.**

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	–1	In all other cases.

Physics

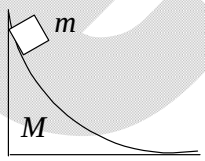
PART – I

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

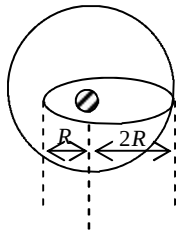
1. In a process on a system, the initial pressure and volume are equal to the final pressure and volume:
   (A) The initial temperature must be equal to the final temperature
   (B) The net heat given to the system in the process must be zero
   (C) The net work done by the system in the process must be zero
   (D) None of these

2. Block placed on wedge of mass M is released. All surface are smooth, then which of following is/are correct:



   (A) C.M. of mass will remain at rest
   (B) Acceleration of centre of mass is in downward direction
   (C) Normal force exerted by floor on wedge will be equal to wedge of weight at some point of its motion.
   (D) Normal force exerted by floor on wedge is always less than combined weight of block and wedge.

3. A spaceship (which has negligible mass) is in an elliptical orbit about a planet. The minimum distance between the spaceship and the planet is R. At the point of maximum distance, spaceship is travelling at a speed v_0,. It then fires his thrusters so that it enters a circular orbit of radius 2R. What is its new speed:



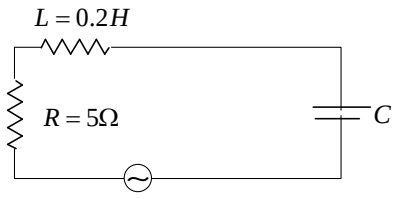
   (A) $v_0\frac{\sqrt{3}}{2}$ (B) $v_0\sqrt{5}$
   (C) $v_0\frac{\sqrt{3}}{5}$ (D) $v_0\sqrt{2}$

4. A nucleus is kept at rest. It breaks into two parts which fly off with velocity in the ratio $27:8$. Then the ratio of radii of the parts is
   (A) 27 : 8 (B) 8 : 27
   (C) 2 : 3 (D) 3 : 2

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

5. An alternating voltage $E = 6\sin 20t + 8\cos 20t$ is applied to series resonant circuit as shown. The correct statements is:



   (A) The capacitance is $12.5mF$
   (B) The resonant current in the circuit is $2A$
   (C) Power dissipated in the circuit is 10 watt
   (D) Power dissipated in the circuit is 20 watt

6. Two non-conducting solid spheres of radii R and 2R, having uniform volume charge densities ρ_1 and ρ_2 respectively are touching each other. The net electric field at a distance 2R from the centre of the smaller sphere, along the line joining the centres of the spheres, is zero then the ratio $\frac{\rho_1}{\rho_2}$ can be:

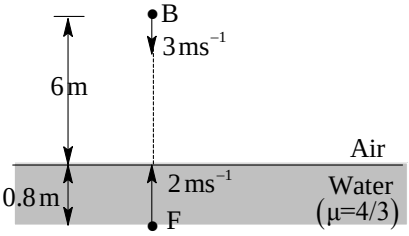
(A) -4 (B) $-\frac{32}{25}$

(C) $+\frac{32}{25}$ (D) 4

7. If one of the slits of a standard Young's double-slit experiment is covered by a thin parallel sided glass slab so that it transmits only one-half the light intensity of the other, then:
(A) The fringe pattern will get shifted toward the covered slit
(B) The fringe pattern will get shifted away from the covered slit
(C) The bright fringes will become less bright and the dark ones will become more bright.
(D) The fringe width will reaming unchanged.

8. According to Bohr's theory of the hydrogen atom, for the electron in the nth allowed orbit:
(A) The linear momentum is proportional to $(1/n)$
(B) The radius is proportional to n
(C) The kinetic energy is proportional to $(1/n^2)$
(D) The angular momentum is proportional to n

9. A fish F, in the pond is at a depth of 0.8 m from the water surface and is moving vertically upward with velocity $2\,ms^{-1}$. At the same instant, a bird B is at a height of 6 m from the water surface and is moving downward with velocity $3\,ms^{-1}$. At this instant, both are on the same vertical line as shown in figure. Which of the following statements are correct?



(A) Height of B, observed by F (from itself), is equal to 5.30 m
(B) Depth of F, observed by B (from itself), is equal to 6.60 m
(C) Velocity of B, observed by F (relative to itself), is equal to $5.50\,ms^{-1}$
(D) Velocity of F, observed by B (relative to itself), is equal to $4.50\,ms^{-1}$

10. n moles of an ideal monatomic gas undergo a process in which the temperature changes with volume as $T=KV^2$. If the temperature of the gas changes from T_0 to $4T_0$ then:

(A) Work done by the gas is $3nRT_0$ (B) Heat supplied to the gas is $6nRT_0$

(C) Work done by the gas is $\frac{3}{2}nRT_0$ (D) Heat supplied to the gas is $\frac{3}{2}nRT_0$

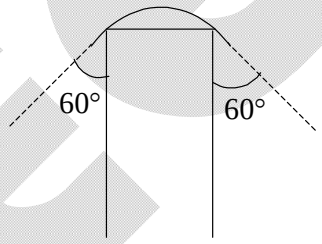
Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

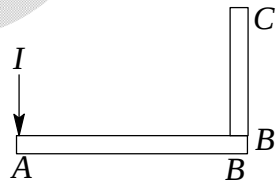
11. Sound of frequency $1000Hz$ from a stationary source is reflected from an object approaching the source at $30\,m/s$, back to stationary observer located at the source. The speed of sound in air is $330\,m/s$. The frequency of the sound heard by the observe is $K\times 400Hz$. Determine $'K'$

12. At an instant, the ratio of the amounts of radioactive substances is $2:1$. If their half lives be 12 & 16 hours respectively, then after 2 days the ratio of amounts of substances left over will be $1:x$ find x:

13. A soap bubble is being blown on a tube of radius 1 cm. The surface tension of the soap solution is $0.05\,Nm^{-1}$ and the bubble makes 60° with the tube as shown in the figure. The excess of pressure over the atmosphere pressure in the tube is 2β Pa. Find the value of β ____.

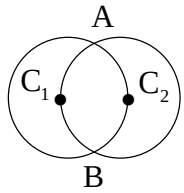


14. A uniform rod AB of mass M is placed in contact with a second rod BC of mass $m=M/2$ on a horizontal smooth table at right angles to each other as shown in figure. Impulse I is imparted to A kinetic energy generated in BC is $\frac{nI^2}{3m}$. Find the value of 'n' ___ .



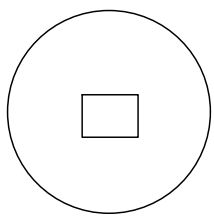
15. A ray of light is incident at an angle of 60° on one face of a prism which has refracting angle of 30°. The ray emerging out of the prism makes an angle of 30° with the incident ray. If the refractive index of the material of the prism is $\mu=\sqrt{a}$, find the value of a ____.

16. Two circular rings of identical radii and resistance of 36Ω each are placed in such a way that they cross each other's centre C_1 and C_2 as shown in figure. Conducting joints are made at intersection point A and B of the rings. An ideal cell of emf 20 V is connected across A and B. Find the power delivered by the cell (in 10^2W) ____.



17. A cubical block of mass M vibrates horizontally with amplitude of 4.0 cm and a frequency of 2.0 Hz. A small block of mass m is placed on the bigger block. In order that the smaller block does not slide on the bigger block, the minimum value of the coefficient of static friction between the two blocks is $(0.16x)$. Find the value of x. (Take $\pi^2=10$ and $g=10\,m/s^2$)

18. Consider the two loops with common centre and on the same plane as shown in the figure. The large circular loop has radius R = 10 cm and the small square loop has side $a=1cm$. There is a current circulating in the large loop given by $i(t)=i_0t/\tau$ with $i_0=\frac{2}{\pi}\times 100A,\ \tau=2\times 10^{-2}$ sec. Emf (in micro volts) induced in the small loop is ____.

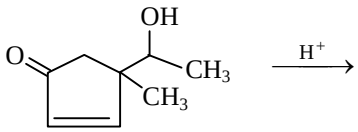


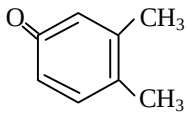
Chemistry

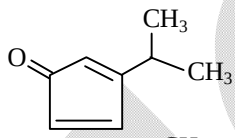
PART – II

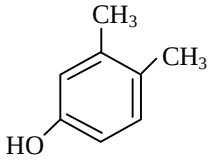
Section – A (Maximum Marks: 12)

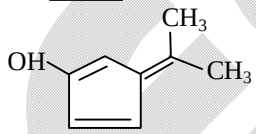
*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

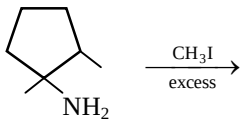
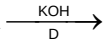
19.  major product is -

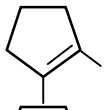
(A) 

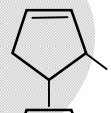
(B) 

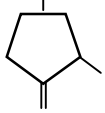
(C) 

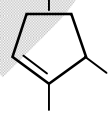
(D) 

20.  A  B then B will be-

(A) 

(B) 

(C) 

(D) 

21. If on a strand of DNA the base sequence is ATTGACGCAT then the sequence transcription on RNA would be -
(A) UAACUGCGUA
(B) AUUCUGCGUA
(C) UAACTGCGUA
(D) TAACTGCGTA

22. The structure of a crystalline lattice AB is rock salt type where B atoms are occupying all the CCP positions but A atoms are occupying all the octahedral voids -
(A) If atoms lying on one tetrad axis of symmetry are removed then molecular formula of the compound become AB
(B) If atoms lying on one triad axis of symmetry are removed then molecular formula of the compound become A_4B_5
(C) If atoms lying on one diad axis of symmetry are removed then molecular formula of the compound become A_5B_8
(D) All of the above molecular formula are correct corresponding to the defects mentioned

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

23. Identify the correct statement (s) regarding the oxidation of $N_2H_5^+$ by $Cr_2O_7^{2-}$ in acid medium.

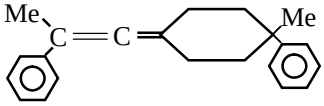
(A) 10 ml 0.2 M dicromate is required for the complete oxidation of 12 ml 0.25 M $N_2H_5^+$

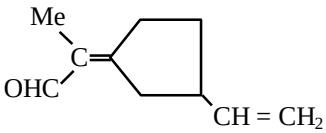
(B) 10 ml 0.2 M dicromate is required for the complete oxidation of 10 ml 0.6 N $N_2H_5^+$

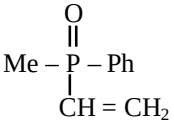
(C) Number of moles of N_2 liberated per mole of dichromate is 1.5

(D) One mole of $N_2H_5^+$ is 3 equivalents

24. Which is resolvable ?

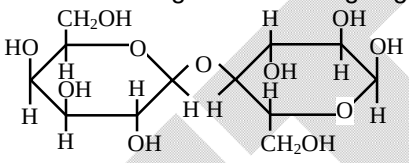
(A) 

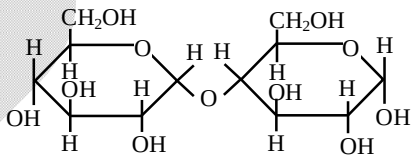
(B) 

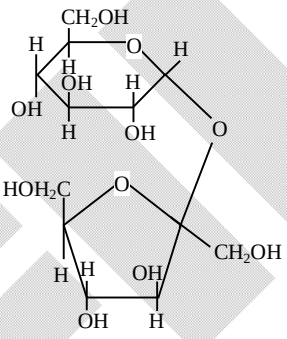
(C) 

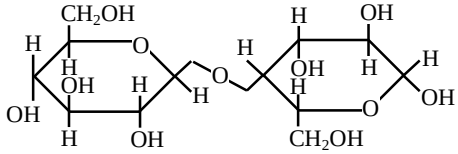
(D) Ethylmethylphenyl amine at pH=2

25. Which of the following is/are reducing sugars -

(A) 

(B) 

(C) 

(D) 

26. Which of the following statements is/are correct?

(A) B_2H_6 reacts with excess NH_3 at low temperature to produce an ionic compound

(B) B_2H_6 reacts with excess of NH_3 at higher temperature to form boron nitride solid

(C) Boron nitride has polymeric layer structure

(D) In borazine there are –ve and +ve charges on B and N respectively

27. The correct products have been reported in reactions –

(A) $CH_2 = CH - CHO \xrightarrow{HCl(g)(-10°C)} CH_2(Cl) - CH_2 - CHO$

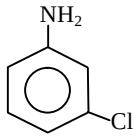
(B) $CH_2 = CH - COOH \xrightarrow{H_2O/H_2SO_4} CH_2(OH) - CH_2 - COOH$

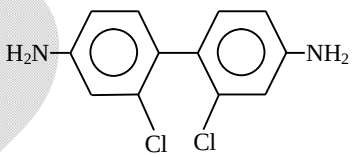
(C) $CH_3 - C(H) = CH - C(=O) - CH_3 \xrightarrow{CH_3OH/H_2SO_4} CH_3 - CH(CH_3) - CH(OCH_3) - C(=O) - CH_3$

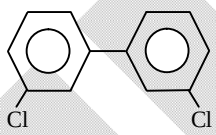
(D) $CH_3 - CH = CH - OCH_3 \xrightarrow{CH_3OH/H^+} CH_3 - CH_2 - CH(OCH_3) - OCH_3$

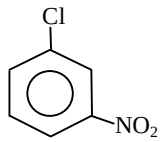
28. $C_6H_6 \xrightarrow[H_2SO_4]{HNO_3} \xrightarrow[Fe]{Cl_2} [X] \xrightarrow[NaOH]{Sn} [Y] \xrightarrow[\Delta]{HCl} \xrightarrow{OH^-} [Z] \xrightarrow[H_3PO_2]{HNO_2} [P]$

The current statement(s) are/is -

(A) Y = 

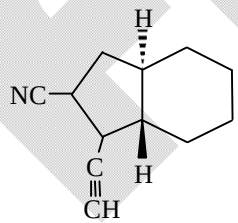
(B) Z = 

(C) P = 

(D) X = 

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

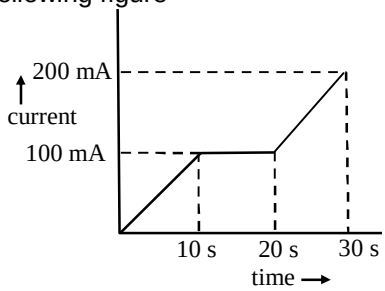
29.  $\xrightarrow[H_2SO_4, H_2S]{HgSO_4}$ gives –'A'.

The degree of unsaturation of compound 'A' is?

30. What is the minimum pH required to prevent the precipitation of ZnS in a solution that is 0.01 M $ZnCl_2$ and saturated with $0.10\,M\ H_2S$?

[Given $K_{sp} = 10^{-21}$, $K_{a_1} \times K_{a_2} = 10^{-20}$]

31. In a Cu-voltameter , mass deposited in 30s is m gm. If the time-current graph is shown in the following figure



The electrochemical equivalent of Cu is $\frac{m}{x}$. Then find the value of 'x' is ___.

32. For a certain reaction the variation of rate constant with temperature is given by the equation

$\ell n\, k_t = \ell n k_0 + \frac{(\ell n 3)t}{10}$ $(t \geq 0^oC)$

The value of temperature coefficient of the reaction is ____.

33. A compound MX_2 has observed and normal molar masses 65.6 and 164 respectively. The apparent degree of ionization of MX_2 is $15x\%$. Then find the value of 'x' is ____.

34. What is the highest oxidation state of nickel in the given complexes of Ni metal?

(i) $Ni(CO)_4$

(ii) K_2NiF_6

(iii) $\left[Ni(NH_3)_6\right](BF_4)_2$

(iv) $K_4\left[Ni(CN)_6\right]$

35. Benitoite is represented as $BaTi\left[Si_3O_n\right]$, the value of n is ……..

36. In order to prepare a buffer of pH= 8.26, the amount of $(NH_4)_2SO_4$ required to be mixed with one litre of 0.1(M) NH_3 (aq) is $0.25x$ mole. The value of 'x' is ____.

(Given:- pK_b of $NH_4OH = 4.74$)

Mathematics

PART – III

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

37. Maximum value of function $f(x)=\left(\sin^{-1}(\sin x)\right)^2-\sin^{-1}(\sin x)$ is:

(A) $\frac{\pi}{4}(\pi+2)$ (B) $\frac{\pi}{4}(\pi-2)$

(C) $\frac{\pi}{4}$ (D) None of these

38. For a parabola $x^2-4xy+4y^2-32x+4y+16=0$, focus is:

(A) $(2,1)$ (B) $(-2,1)$

(C) $(-2,-1)$ (D) $(2,-1)$

39. Given three non – zero, non – coplanar vectors, and $\vec{a},\vec{b}$ and $\vec{c}$ and $\vec{r}_1=p\vec{a}+q\vec{b}+\vec{c}$ and $\vec{r}_2=\vec{a}+p\vec{b}+q\vec{c}$ if the vectors $\vec{r}_1+2\vec{r}_2$ and $2\vec{r}_1+\vec{r}_2$ are collinear then (p, q) is

(A) (0, 0) (B) (1, –1)

(C) (–1, 1) (D) (1, 1)

40. Solution of differential equation $f'(x)\,f'''(x)=3\left(f''(x)\right)^2,\left(y=f(x)\right)$ is:

(A) $x=k_1y^2+k_2y+k_3$ (B) $y=k_1x^2+k_2+k_3$

(C) $x=\sin y$ (D) None of these

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

41. Let $f(x)=\begin{cases}\lim\limits_{n\to\infty}\left(\frac{px}{n}\sum\limits_{r=1}^{n}\frac{\left[r^2-e^{-x}+r-1\right]}{r(r+1)}\right)+\lambda, & x>0\\ q, & x=0\\ \lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\frac{\left\{r^2+r+e^{x}-1\right\}}{r(r+1)}, & x<0\end{cases}$

Is differentiable in $\mathbb{R}$:

$\left([.]\text{is G.I.F. and }\{.\}\text{ is F.P. of x}\right)$ Then:

(A) p=1 (B) q=1

(C) $p+q+\lambda=3$ (D) if g is inverse of F then $g'(1/2)=2$

42. Consider $f(x)=4x^4-24x^3+31x^2+6x-8$ be a polynomial function and $\alpha,\beta,\gamma,\sigma$ are the roots. $(\alpha<\beta<\gamma<\sigma)$. So:

(A) $\sigma^{\beta}+\frac{1}{\sigma^{\alpha}}+\sigma^{\gamma}+\gamma^{\sigma}=36$

(B) $\sigma^{\beta}+\frac{1}{\sigma^{\alpha}}+\sigma^{\gamma}+\gamma^{\sigma}=18$

(C) $\int_{2\alpha}^{2\beta}\frac{x^{\sigma+1}-5x^{\gamma+1}+2\beta|x|+1}{x^2+4\beta|x|+1}dx=2\ln 2$

(D) $\int_{2\alpha}^{2\beta}\frac{x^{\sigma+1}-5x^{\gamma+1}+2\beta|x|+1}{x^2+4\beta|x|+1}dx=\ln 2$

43. Consider the equation $z^2-(3+i)z+(m+2i)=0(m\in R)$. If the equation has exactly one real and one-non-real complex root, then which of the following hold(s) good:

(A) Modulus of the non-real complex root is 2

(B) The value of m is 3

(C) Additive inverse of non-real root is $(-1-i)$

(D) Product of real root and imaginary part of non-real complex root is 2

44. $\sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\frac{m^2n}{\left(n.3^m+m.3^n\right)}=\frac{p}{q}$ (Where $p\,\&\,q$ are coprime) then:

(A) $p+q=41$

(B) $p<q$

(C) p is perfect square

(D) q is perfect square

45. Let $p(x)=4\sin^3 x-\sin x+2\left(\sin\frac{x}{2}-\cos\frac{x}{2}\right)^2$, then which of the following is/are correct?

(A) Range of $p(x)$ is $[1,3]$

(B) Range of $p(x)$ is $[0,4]$

(C) Number of solution of $p(x)=1$ in $[0,\pi]$ is 2

(D) Number of solution of $p(x)=1$ in $[0,2\pi]$ is 4

46. Let m,n be real numbers satisfying the relation:

$$\log_2\left(m^2+n^2+1\right)+\log_3\left(\frac{1+2\sin^2\left(\frac{2\pi}{7}\right)}{2-\cos\left(\frac{4\pi}{7}\right)}\right)=\log_2 n+\log_2(2m+2-n)$$

Identify which of the following statement(s) is/are correct?

(A) $m+n=2$

(B) $\big||m|-|n|\big|=4$

(C) Area of the figure enclosed by $|x|+|y|=|m|+|n|$ is 8 square units.

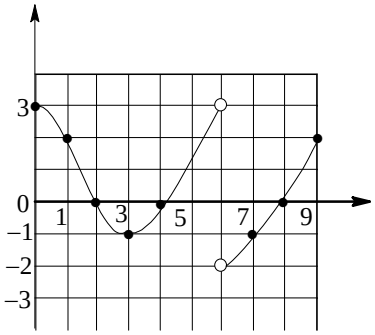
(D) Area of the figure enclosed by $|x|+|y|=|m|+|n|$ is 16 square units.

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

47. A and B are two square matrices such that $A^2B = BA$ and if $(AB)^{10} = A^kB^{10}$ then the value of $k - 1020$ is....

48. The graph of the derivative f of a continous function f is shown with f(0) = 0



If $f(x)$ has local minima at $x = x_1$ and $x = x_2$; f has inflection point at $x = k$; number of critical points of $y = f(x)$ is w then $x_1 + x_2 - k - w$ is.....

49. Number of values of $x \in [0, \pi]$ where $f(x) = [4\sin x - 7]$ is non derivable is ($[.]$ is G. I. F)

50. If the equation on reflection of $\frac{(x-4)^2}{16} + \frac{(y-3)^2}{9} = 1$ about the line $x - y - 2 = 0$ is $16x^2 + 9y^2 + k_1x - 36y + k_2 = 0$ then $\frac{k_1 + k_2}{22}$ is

51. If $0 < \alpha < 30°$, then find the value of $\log_2\left(\frac{8\cos(30° - 2\alpha) - \frac{8}{\cot\alpha + \cot(30° - \alpha)}}{\sqrt{3}}\right)$

52. Find the value of $\log_{\sqrt{5}}\left(\frac{\sqrt{(5\sqrt{5}+5)\sqrt{(5\sqrt{5}+5)^2\sqrt{(5\sqrt{5}+5)^3\sqrt{....}}}}}{(6+2\sqrt{5})\sqrt[3]{215 - 18\sqrt[3]{215 - 18\sqrt[3]{215 - 18\sqrt[3]{....}}}}}\right)$

53. Three circles C_1, C_2, C_3 with radii r_1, r_2, r_3 ($r_1 < r_2 < r_3$) respectively are given as $r_1 = 2$, and $r_3 = 8$ they are placed such that C_2 lies to the right of C_1 and touches it externally, C_3 lies to the right of C_2 and touches it externally. There exist two straight lines each of which is a direct common tangent simultaneously to all the three circles then r_2 is equal to __.

54. If K^{th} term is the greatest term in the square $a_n = \frac{n}{70 + n^2}, n \in N$, then k is____